

Brock Allan

Software Engineer | Machine Learning Engineer | Game Designer

Phone	Address	Email	LinkedIn	Portfolio
(951) 655-0133	Moreno Valley, CA	brockallan55@gmail.com	LinkedIn-BrockAllan	Portfolio

To whomever it may concern,

I am writing to express my interest in the position for which I am applying to your company. Throughout my academic career, I have developed a strong foundation in both theoretical computer science and practical software development. My background spans building systems-level architecture, researching generative AI applications, and evaluating network security, making me well-equipped to tackle complex technical challenges.

My recent work focused heavily on applied machine learning and computer vision. My recent thesis evaluated the implementation of AI and prompt engineering within game development, specifically analyzing procedural generation tools and AI-assisted pair programming. Additionally, I have hands-on experience designing and debugging neural network architectures, including CNN layers, spatial pooling, and diffusion models. I thrive on translating theoretical models into functioning, optimized code.

Beyond artificial intelligence, I have a deep appreciation for robust systems engineering and cybersecurity. I have developed custom software tools from the ground up, including an inverted index search engine, a Unix-like shell interpreter, and operating system scheduling simulations. This systems-level perspective is complemented by practical lab experience in web security, network analysis using Wireshark, and binary exploitation via GDB, allowing me to approach software development with a security-first mindset.

I am drawn to your company because I believe it would provide me the experience and room to grow in this industry. I would welcome the opportunity to bring my interdisciplinary technical background and passion for rigorous problem-solving to your engineering team.

Thank you for your time and consideration. I look forward to the possibility of discussing how my skills align with your needs].

Sincerely,

Brock Allan